Readiness Scores: What Wearables Measure, What Recovery Science Suggests, and How a Readiness Score Should Be Built

A readiness score is not a fitness score. It is trying to answer "how prepared is my body for training today?". This makes it a short-term estimate built from recent sleep, autonomic signals (the involuntary nervous-system activity that regulates things like heart rate), recent training demand and signs that the body may be dealing with something else.

Apple's Vitals app is a useful example of the general logic. It looks for overnight metrics that move outside the user's own typical range and places Training Load beside those signals so the user can interpret recovery in context. [1]

We need to understand what providers do at the moment so we can compare these approaches with the scientific research and derive a useful algorithm. The broad market pattern is clear even where the exact calculations are hidden.

WHOOP's public descriptions connect recovery with heart rate variability (HRV), resting heart rate (RHR), respiratory rate and sleep. It also makes training demand part of the next day's recovery and sleep need. More recently WHOOP has expanded strain to include muscular load alongside cardio load, which matters because a readiness system is only as good as the load model it reacts to. [2]

Garmin has taken one of the most explicit multi-input approaches. Garmin describes Training Readiness as a 1-100 score combining sleep, recovery time, HRV, acute load and stress, while Body Battery acts as a day-long energy estimate driven by HRV, stress, sleep and activity. The important choice is that overnight recovery and recent training history are interpreted together. [3]

Fitbit's Daily Readiness Score has used sleep, recent activity, HRV and RHR to guide whether the user should push or recover. Google's more recent work has placed greater emphasis on Cardio Load and personalised targets. This points towards recovery being interpreted beside a clearer load model rather than asking one number to describe everything. [4]

The science takes us in the same general direction. Readiness should be based on individual baselines and recent trends. Universal thresholds are still useful for safety and plausibility, although they are a poor description of a person's normal recovery state.

This is most clear with HRV. Athlete monitoring research consistently treats HRV as a personal and longitudinal measure. A single unusual morning can be noise. A sustained change across several comparable measurements carries more weight. HRV is also nonspecific: training, short sleep, illness, alcohol, travel and psychological stress can all change it. [5] [7] [8] [9]

There is an important measurement caveat here. A 2025 study found that pulse rate variability and heart rate variability were not interchangeable during posture transitions, with disagreement increasing away from stable resting conditions. This matters because many wearables estimate HRV like signals from PPG (photoplethysmography, the optical sensor that shines light into the skin) rather than ECG (electrocardiography, which measures the heart's electrical activity and is generally treated as the reference method). Overnight or controlled resting measurements are therefore the most useful autonomic inputs for readiness. [6]

Several days are more informative than one. Plews and colleagues found that averaging at least 3 valid LnRMSSD (a mathematically transformed measure of HRV that makes day-to-day comparisons more reliable) measurements in a week produced a much more stable assessment than depending on one reading. This does not mean the latest night should disappear into a weekly average, because readiness is still about today. It means the latest measurement should be interpreted beside a short recent trend. [10]

The same caution applies when HRV is used as a stand-in for every kind of recovery. In a small resistance training study, the recovery of HRV and squat bar velocity did not follow the same time course over 72 hours. Other work has found HRV can recover while neuromuscular performance and soreness are still impaired. A normal autonomic signal cannot be used to declare a trained muscle recovered. [11] [12]

This gives us 3 main blocks for the readiness calculation: sleep recovery, autonomic recovery and recent load recovery. Illness-like signals and explicit context sit around that core as modifiers. Data confidence sits beside the result.

What the readiness score is estimating

The score should estimate general readiness for physical strain during the coming day. A high score means the measured recovery signals look favourable for that person. A low score means one or more recovery systems are showing meaningful strain.

General readiness also has limits. A person may be systemically well recovered while one muscle group is still sore or carrying a large residual load. The main number should be accompanied by channel level explanations and local muscular warnings where that information exists. This is more honest than forcing every form of readiness into the same value.

The inputs are calculated elsewhere. Personal Metric Baselines defines the comparable measurement windows, centre, spread, exclusions and confidence phases for each metric. Sleep Need, Debt, Recovery and Stages defines core sleep need and the persistent sleep-burden calculation. Sleep Scores defines duration and continuity components. Strain Scores and Resistance Training and Muscular Load define cardiovascular, whole-body muscular and muscle-group load. Readiness consumes those outputs and does not recalculate them differently.

The core calculation

The starting score is:

$$\text{core readiness} = 0.35 \times \text{sleep recovery} + 0.35 \times \text{autonomic recovery} + 0.30 \times \text{recent load recovery}$$

All 3 components run from 0 to 100. The weights are implementation choices supported by the broad order of importance in the literature. They should be stored as configurable values and tested against future outcome data. Confidence is excluded from this equation.

The 35/35/30 split gives sleep and autonomic state equal influence while keeping enough space for training demand to change the interpretation. It also avoids giving normal respiratory rate or the absence of an illness tag a permanent block of positive points. Those signals become useful when something changes.

Calculating sleep recovery

Readiness should use the parts of sleep that most directly describe recovery: how much sleep occurred, how fragmented it was and how much sleep burden has accumulated. It should not use the full sleep score. Doing that would pull timing, regularity, stages and overnight physiology into readiness, then count some of the same physiology again in the autonomic component.

Let D be the duration component from Sleep Scores, C the continuity component, T the current sleep burden in minutes from Sleep Need, Debt, Recovery and Stages, Q the shortfall created by the latest night and N the person's core sleep need in minutes. Duration already reflects the latest shortfall, so the burden input removes it before scoring:

$$B = \text{max}(0, T - Q)$$

This leaves the burden carried into the night after any recovery sleep has been applied. It stops the latest missing hour being charged once through duration and then immediately charged again through debt.

$$\text{burden nights} = \frac{B}{N}$$

The burden score is linearly interpolated between these points:

- 0 burden nights $=$ 100
- 0.5 burden nights $=$ 90
- 1 burden night $=$ 75
- 2 burden nights $=$ 50
- 3 burden nights $=$ 25
- 4 or more burden nights $=$ 0

Sleep recovery is then:

$$\text{sleep recovery} = 0.55D + 0.25C + 0.20\,\text{burden score}$$

Duration has the largest share because acute sleep loss has a consistent negative effect on physical performance and perceived effort. Continuity captures a poor night that total minutes alone can hide. Burden makes repeated restriction matter without allowing one short night to rewrite the person's sleep need.

Duration and burden are required. If continuity cannot be calculated, the duration and burden weights are rescaled within this component:

$$\text{sleep recovery} = \frac{0.55D + 0.20\,\text{burden score}}{0.75}$$

This lowers confidence. It does not turn missing awakenings into perfect continuity.

The severe short-sleep rules from Sleep Scores also need to carry through to readiness. A duration shortfall of 3 hours caps readiness at 70, 4 hours caps it at 50 and 5 hours or more caps it at 30. Severe fragmentation, defined in Sleep Scores, caps readiness at 70. These caps stop strong HRV or a quiet training week from hiding a night that is very unlikely to support normal performance.

Calculating autonomic recovery

Autonomic recovery should use overnight or controlled resting data from a consistent device and method. RMSSD is the preferred HRV input because it describes short term successive beat variation and has better statistical properties for this purpose. Apple Health reports HRV as SDNN, the standard deviation of normal-to-normal intervals, so the calculation also needs a defined SDNN path. RHR should come from the same sleep or resting window where possible. The metric specific baseline work is handled by Personal Metric Baselines. [18] [20]

RMSSD and SDNN should be treated as separate measurement types. SDNN describes the overall variability present during the recorded period and changes materially with recording length. A short Apple Watch SDNN value cannot share a reference with a 5 minute, overnight or 24 hour SDNN value, and SDNN should not be converted into an estimated RMSSD. The selected HRV series must keep the metric, provider or device, sampling context and recording method consistent. Where recording duration is unavailable, the provider, device and collection method define the series. [18] [19]

Use RMSSD when a valid comparable RMSSD is available. If the source only supplies SDNN, use SDNN as the fallback. Where both are available, score RMSSD only so the same physiological signal does not enter autonomic recovery twice. Other time domain measures should remain unscored until they have their own measurement definition and personal reference.

For Apple Watch data:

$$\text{nightly SDNN} = \text{median}(\text{valid SDNN samples within the main sleep window})$$

If only one valid sample exists within the window, that sample becomes the nightly SDNN. Daytime opportunistic samples should not be mixed into the overnight series. A controlled resting series can be used separately when it is collected under comparable conditions each time.

The HRV input is then:

$$\text{HRV input} = \ln(\text{RMSSD}) \quad \text{when RMSSD is selected}$$

$$\text{HRV input} = \ln(\text{SDNN}) \quad \text{when SDNN is selected}$$

Both calculations use the natural logarithm of the value in milliseconds. The centre and spread must belong to the selected metric-specific series.

For each valid night, calculate a standardised deviation from the personal reference:

$$z_{\text{HRV}} = \frac{\text{HRV input} - \text{HRV centre}}{\text{HRV spread}}$$

$$z_{\text{RHR}} = \frac{\text{RHR} - \text{RHR centre}}{\text{RHR spread}}$$

Using SDNN as a valid fallback does not lower the readiness value or its confidence by itself. Confidence falls when the measurement method is inconsistent, the sampling window is unsuitable or the metric-specific reference does not contain enough comparable days. These rules follow Personal Metric Baselines.

The current night needs to remain visible, although the recent trend should carry slightly more weight. Let recent z be the median of the latest 3 valid comparable nights, including the current night. Those readings should normally fall within the last 4 calendar nights. If fewer than 3 are available, use the available readings and lower confidence.

$$\text{HRV adverse deviation} = \text{max}(0, -(0.40 z_{\text{HRV,current}} + 0.60 z_{\text{HRV,recent}}))$$

$$\text{RHR adverse deviation} = \text{max}(0, 0.40 z_{\text{RHR,current}} + 0.60 z_{\text{RHR,recent}})$$

The signs differ because lower HRV and higher RHR are the usual adverse directions. Values in the favourable direction stop at 0 adverse deviation and receive no bonus above 100. A very high HRV value can still be shown as unusual, particularly when it is sustained, but it should not automatically add readiness points because both increases and decreases have been observed during poor adaptation. [7]

Each adverse deviation is converted to a component score using linear interpolation:

- 0 to 0.5 spreads $=$ 100
- 1 spread $=$ 85
- 1.5 spreads $=$ 65
- 2 spreads $=$ 40
- 3 spreads $=$ 10
- 4 or more spreads $=$ 0

When both metrics are available:

$$\text{autonomic recovery} = 0.60\,\text{HRV score} + 0.40\,\text{RHR score}$$

When only one is available, use that score and lower confidence. At least one valid autonomic metric is required. There is no separate trend penalty because the current and recent measurements are already combined in the adverse deviation. HRV and RHR are not charged again as ordinary anomaly penalties later in the calculation.

Calculating recent load recovery

Recent load recovery asks how much training demand is likely to stay relevant today. The cardiovascular and muscular channels stay separate until the end. Their calculation comes from the Strain Scores and Resistance Training and Muscular Load papers.

A fixed 7-day to 28-day ratio is not suitable here. Acute:chronic workload ratios (recent training load compared with your usual training load) have statistical and causal limitations, and fixed injury risk zones are not supported strongly enough to become readiness thresholds. [13] We can still compare recent demand with personal history, using a decayed exposure and the same baseline logic as the other metrics.

Let C1, C2 and C3 be cardiovascular load from 1, 2 and 3 days ago. Let M1, M2 and M3 be full body muscular load over the same days.

$$\text{cardio residual exposure} = 0.60C_1 + 0.30C_2 + 0.10C_3$$

$$\text{muscular residual exposure} = 0.50M_1 + 0.30M_2 + 0.20M_3$$

The slightly slower muscular decay reflects the common 24-72 hour time course of neuromuscular impairment and soreness after demanding resistance exercise. It remains a general approximation. Exercise selection, eccentric loading, training experience and the muscle involved can all change recovery time.

Each residual exposure is compared with its own 28-day personal reference, extending to 60 days when the valid history is sparse. Let z exposure be:

$$z_{\text{exposure}} = \frac{\text{residual exposure} - \text{residual exposure centre}}{\text{residual-exposure spread}}$$

Only exposure above the personal centre reduces the component:

$$\text{adverse exposure} = \text{max}(0,\ z_{\text{exposure}})$$

Convert each channel using linear interpolation:

- 0 to 0.5 spreads = 100
- 1 spread = 90
- 2 spreads = 65
- 3 spreads = 35
- 4 or more spreads = 10

If both channels are valid, the more affected channel receives 70% of the combined component and the better channel receives 30%:

$$\text{recent load recovery} = 0.70\,\text{lower channel score} + 0.30\,\text{higher channel score}$$

If only 1 channel is measured reliably, use it and lower confidence. A reliably observed rest period gives a load recovery score of 100. This component describes residual demand rather than whether the person has trained enough. The missing load remains unknown.

For resistance training, apply the muscular residual equation to each muscle group as well. A muscle group adverse exposure above 2 spreads creates a “reduced local readiness” warning; above 3 spreads creates a “very reduced local readiness” warning. These warnings do not apply another subtraction to the main score. They explain where the muscular component came from and stop a whole body number from implying that every body part is equally recovered.

Illness-like signals and concordance

Respiratory rate, oxygen saturation and skin temperature are useful because they can move during infection and other physiological disturbance. Prospective wearable studies have detected infection related changes across overnight RHR, HRV, respiratory rate and skin temperature, although these systems are screening signals and not diagnoses. [16] [17]

The measurements are correlated, so the algorithm should work with groups:

- Autonomic group: the autonomic recovery component is below 70.
- Respiratory group: respiratory rate is above its metric specific alert boundary, oxygen saturation is below its personal lower boundary, or both move adversely together.
- Temperature group: overnight skin-temperature variation crosses its metric specific alert boundary.

A group becomes persistent when one of its metrics crosses the alert boundary on 2 consecutive valid nights. A group can become current-night concordant when 2 related measurements cross together. 1 mild deviation on one night remains an explanation and does not change the score.

The anomaly modifier is applied after the core score:

- No persistent non-autonomic group $=$ 0-point penalty.
- 1 persistent respiratory or temperature group $=$ 5-point penalty.
- 1 persistent non-autonomic group plus an impaired autonomic group = 10-point penalty and a readiness cap of 75.
- Both respiratory and temperature groups persist $=$ 12-point penalty.
- Respiratory and temperature groups persistent with impaired autonomic recovery = 18-point penalty and a readiness cap of 60.

$$\text{final readiness} = \text{round}(\text{clamp}(\text{core readiness} - \text{anomaly penalty}, 0, 100))$$

After this calculation, apply the lowest active cap. The published value is the resulting whole number from 0-100.

This uses autonomic disturbance as confirmation without subtracting separate points for low HRV and high RHR again. A severe absolute oxygen-saturation result or a provider quality/safety warning uses the absolute handling defined in Personal Metric Baselines, caps readiness at 40 and shows a health warning. The score will never present this as a diagnosis.

Subjective context

Subjective monitoring deserves a place because perceived fatigue, soreness, stress, mood and sleep quality can react to training even when objective measures are unchanged. Reviews have found these measures can be sensitive, although their relationship with training load varies and their usefulness depends on consistent questions and meaningful outcomes. [14] [15]

Until the product has a short, consistently administered morning journal, most context should explain the result rather than alter it. Alcohol, travel, altitude, psychological stress and menstrual cycle context can help explain a physiological change. They should not receive generic deductions simply because a tag exists.

An explicit "I feel ill" or "I am unwell" response caps readiness at 40. Fever, chest pain, severe breathlessness or another urgent symptom should replace training guidance with an appropriate safety message. Local muscle soreness should attach to the relevant muscle warning. Widespread soreness can support the muscular-load explanation but is not subtracted for a second time.

If a morning questionnaire is introduced later, fatigue, soreness, stress, mood and perceived recovery should use the same scale every day and be validated against meaningful outcomes before being added to the arithmetic. This keeps a useful future input open without pretending that an unvalidated collection of tags is a precise physiological component.

Missing data and confidence

Confidence should sit beside readiness. The score requires:

- A valid main sleep duration and a calculable sleep burden.
- At least 1 valid overnight or controlled resting autonomic metric.
- Reliable activity coverage for the previous 3 days.
- At least 7 comparable days for the autonomic and residual-load references.

If these minimums are not met, the system should show the available components and a calibration state rather than a complete readiness number. This avoids silently redistributing a missing 35% autonomic block into sleep or treating an unknown training day as rest.

High confidence requires valid duration and continuity, both HRV and RHR, reliable cardiovascular and muscular coverage, a stable provider/device and at least 28 comparable days for the personal references. Moderate confidence has the required data and 14-27 comparable days, or one supporting input is missing. Low confidence has 7-13 comparable days, only one autonomic input or one reliable load channel. Missing optional respiratory or temperature data lowers anomaly confidence and does not lower readiness.

A device or provider change should move the affected metric back into calibration as defined in Personal Metric Baselines. Historical values from a materially different method should not be mixed into the same reference just to preserve a high confidence label.

Interpreting the score

The following bands make the number generally easier to communicate and understand:

- 85-100: high readiness. Recovery signals are favourable and there is no strong modifier.
- 70-84: good readiness. Some strain is present, although normal training is likely reasonable.

- 55-69: moderate readiness. One important system is meaningfully affected; reduce or adapt demand.
- 40-54: low readiness. Recovery is clearly incomplete or several signals agree.
- 0-39: very low readiness. Strong recovery or illness-like concerns are present.

These labels should be presented with the component explanation. A score of 62 caused by short sleep means something different from 62 caused by heavy leg training or concordant respiratory and temperature changes. The number guides attention; the explanation guides action.

Validation

The weighting and interpolation values above are initial design choices. Readiness is not a directly measured biological quantity, so the algorithm needs a declared validation target. The useful targets are next day perceived recovery, planned session completion, session RPE relative to expected effort and simple performance measures where they are available.

Each daily record should store the raw inputs, baseline centres and spreads, current and recent deviations, component scores, active caps, modifier flags, confidence and the final value. This makes later calibration possible and prevents a changed coefficient from becoming impossible to audit.

Validation should begin with prospective observation. Compare the score and its components with next-day outcomes without changing the user's training plan. Check calibration separately for resistance, endurance, mixed training, sex, age and device source where sample size allows. Only then should weights, decay rates and caps be adjusted. The score should be judged on calibration and useful discrimination rather than whether its average looks similar to another provider's score.

Conclusion

Taken together, the readiness calculation supported by this research has the following form:

- Sleep recovery accounts for 35%. It combines duration 55%, continuity 25% and persistent sleep burden 20%. Duration and burden are required.
- Autonomic recovery accounts for 35%. It uses 60% HRV and 40% RHR when both are available, combining the current night 40% with the latest 3-night trend 60%. HRV uses LnRMSSD where available or a separate source-specific LnSDNN series when SDNN is the only valid input.
- Recent load recovery accounts for 30%. Cardiovascular load decays across 3 days at 60/30/10 and muscular load at 50/30/20. The more affected channel carries 70% of the combined load component.
- All personal deviations use the metric-specific centres, spreads, exclusions and confidence rules in Personal Metric Baselines.

- The core score is 0.35 x sleep recovery + 0.35 x autonomic recovery + 0.30 x recent load recovery.
- Respiratory and temperature groups act as post-score modifiers. Concordance with impaired autonomic recovery can cap readiness at 75 or 60.
- Explicit illness or a severe oxygen-saturation warning caps readiness at 40 and produces a health warning rather than a diagnosis.
- Short-sleep caps are 70 after a 3-hour shortfall, 50 after 4 hours and 30 after 5 hours or more. Severe fragmentation caps readiness at 70.
- Local muscular residuals above 2 and 3 personal spreads create reduced and very reduced muscle-group readiness warnings without another global deduction.
- Confidence is reported separately. A score needs sleep, at least one autonomic input, reliable 3-day activity coverage and at least 7 comparable reference days.

This creates a readiness score that reacts to the main recovery systems while keeping their meanings clear. Sleep stages do not quietly become training readiness, HRV does not claim to measure muscle recovery, and several correlated illness signals do not each collect their own full penalty.

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